

Verify truth tables of AND, OR, NOT, NOR, NAND, XOR, XNOR gates & simplifications of Boolean gates.

★ Equipments Required:-

- (I) Digital trainer circuit
- (II) Power Supply
- (III) Patch cords

★ Theory:-

a) NOT GATE:-

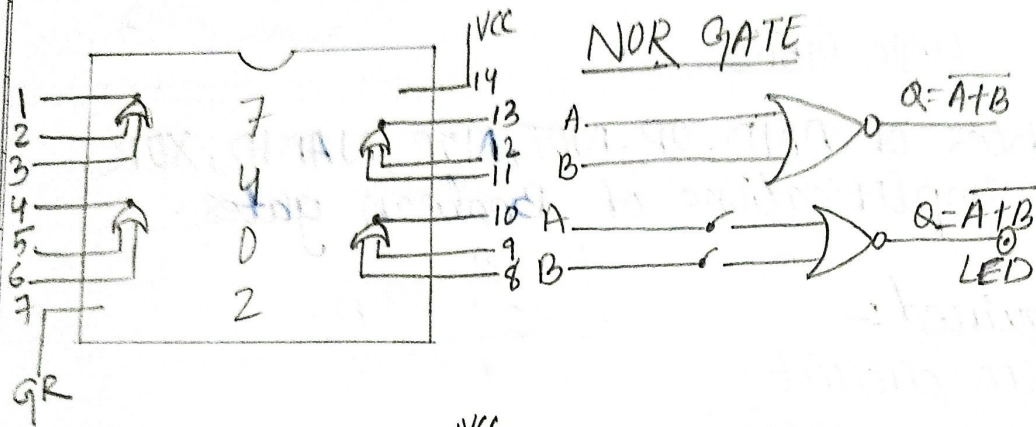
- A NOT gate is also called as inverter which has only one input and one output.
- It is a device whose output is always the complement of its input.
- The output of a NOT gate is the logic 1 state when its input is 1 state.
- Mathematical expression: $F = \bar{A}$

b) AND GATE:-

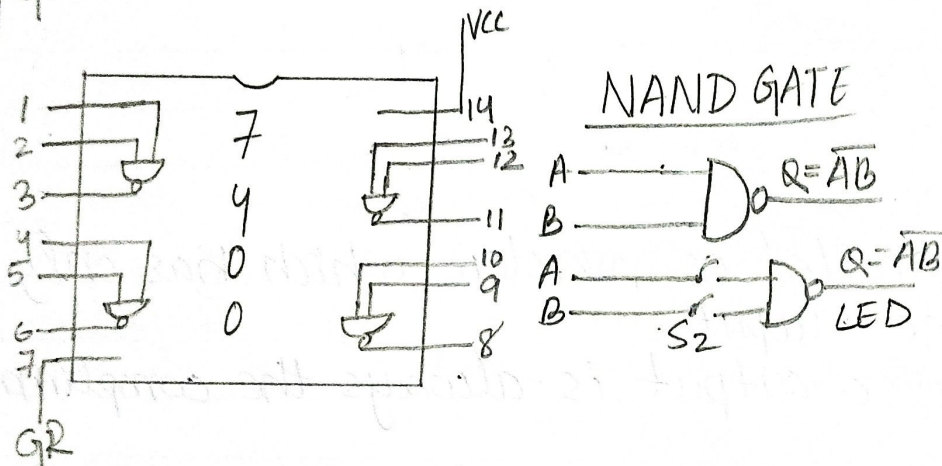
- An AND gate has two or more inputs but only one output.
- The output is logic 1 state only when each of its inputs is at logic 1 state.
- The AND gate is a logic circuit whose output is high when all inputs are high.
- Its IC number is 7408 and it has 4 Nos of AND gate.
- Mathematical expression: $F = AB$

c) OR GATE:-

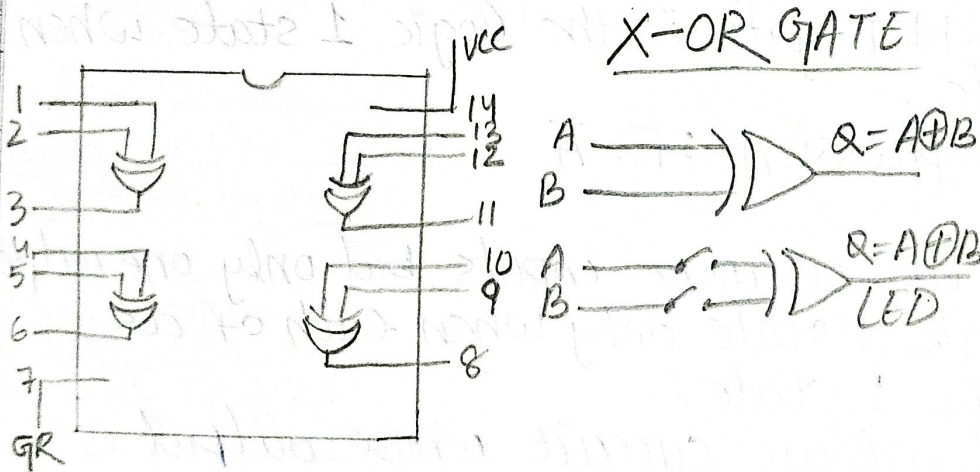
- An OR gate may have two or more no. of input but only one output.
- The output is logic 1 state even if one of its inputs is in logic 1 state.



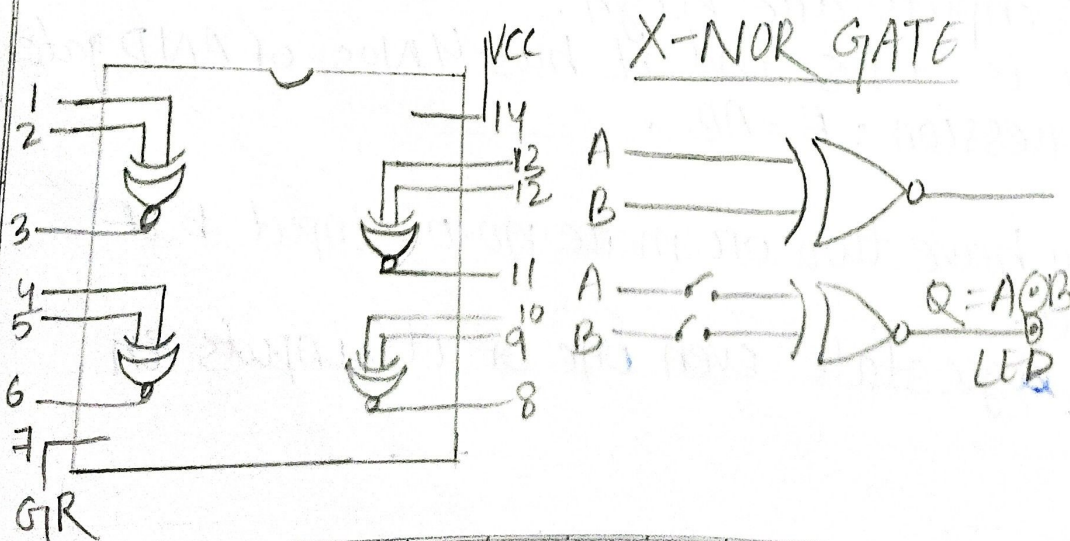
I/P		O/P
A	B	$\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0



I/P		O/P
A	B	$\overline{AB}$
0	0	1
0	1	1
1	0	1
1	1	0



I/P		O/P
A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0



I/P		O/P
A	B	$A \odot B$
0	0	1
0	1	0
1	0	0
1	1	1

→ It's IC number is 7432 and it has 4 Nos of AND gate.

→ Mathematical expression: $F = A + B$.

d) NOR GATE:-

→ NOR gate means the combination of an OR gate and NOT gate i.e. the output of OR gate is feed to NOT gate.

→ The output is logic 1 only when all of inputs is logic and for other combination of inputs, the output is logic 0 level.

→ An NOR gate is a logic circuit whose output is low when any one input is high.

e) NAND GATE:-

→ NAND gate is combination of AND gate and NOT gate.

→ The output of the AND gate is feed in not gate.

→ The output is logic 0 taken each of the input is logic 1 and for any other combination of inputs, the output logic is 1.

→ It's IC number is 7400 & it has 4 Nos of NAND gates.

→ Mathematical expression: $F = \overline{AB}$.

F) X-OR GATE:-

→ An X-OR gate is a also two input and one output logic circuit.

→ The output is 1, when one and only one of two input is 1.

→ When both inputs are zero (0) or both input are one (1) the output is 0.

→ It's IC number is 7486 & it contains 4 Nos of X-OR gates.

→ Mathematical expression: $F = \overline{A}B + A\overline{B} = A \oplus B$.

g) X-NOR GATE:-

→ An XNOR gate is the combination of an X-OR gate and NOT gate.

→ The output is 1, when one and only one of the two input is 1.

→ The output is logic 1, only when both the input are 0 or when both input are 1.

→ Its IC number is 74266 and it contains 4 Nos of 7 NOR gate.

→ Mathematical expression: ~~$AB + \bar{A} + B$~~ $AB + \bar{A}B = A \odot B$

(I) Connect the patch cards as per the circuit diagram.

(II) Give input through switches.

(III) Verify truth table according to lead LED (Light Emitting Diode).

Conclusion:-

From the above experiment we have studied and verified the truth table of different logic gates.

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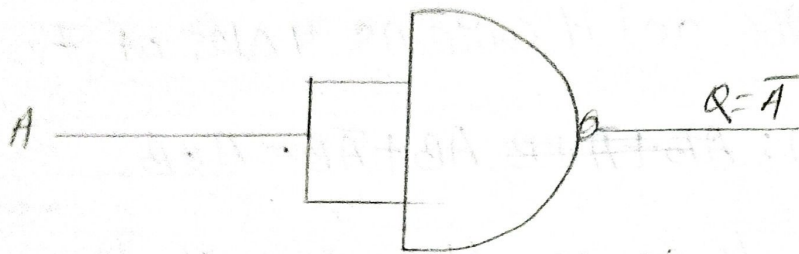
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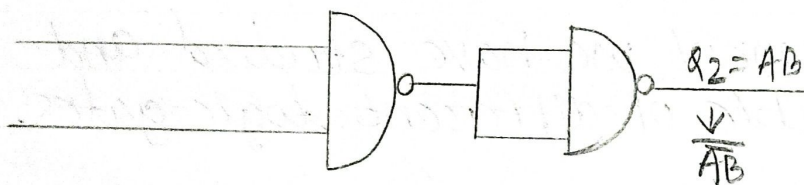
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NAND Gate as NOT Gate



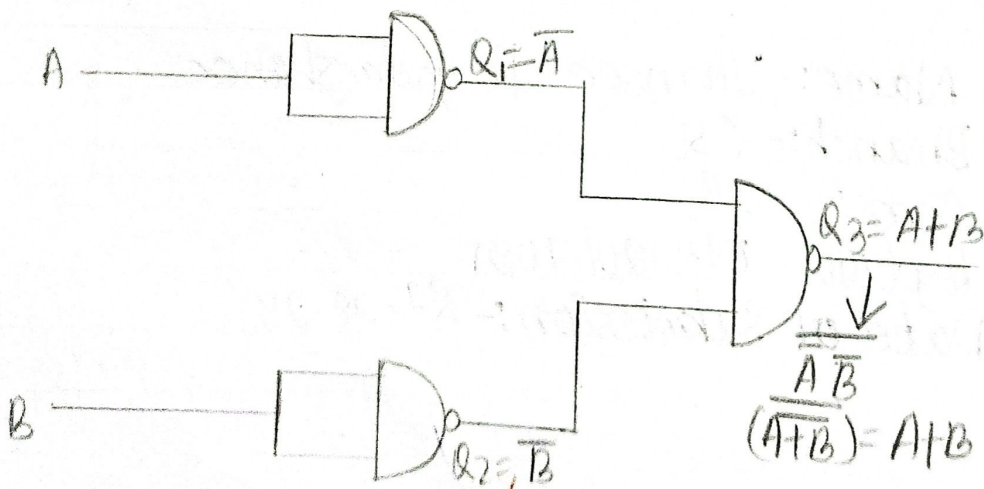
I/P	O/P
A	$\bar{A}$
0	1
1	0

NAND Gate as AND Gate



I/P	$\overline{AB}$	$\overline{\overline{AB}} = AB$	
A	B	Q_1	Q_2
0	0	1	0
0	1	1	0
1	0	1	0
1	1	0	1

NAND Gate as OR Gate



I/P	$\bar{A}$	$\bar{B}$	$(\overline{A+B}) = A \oplus B$	
A	B	Q_1	Q_2	Q_3
0	0	1	1	0
0	1	1	0	1
1	0	0	1	1
1	1	0	0	1